实验报告

<table>
<tr><td colspan="4">实验报告</td></tr>
<tr><td>题目</td><td colspan="3">实验三 SQL-查询</td></tr>
<tr><td>姓名</td><td>刘师妤</td><td>学号</td><td>2024201568</td></tr>
<tr><td colspan="4">实验环境（计算机配置，操作系统，编程语言，编程工具等）
硬件平台：MacBook Air（M5，24G）
操作系统：macOS
DBMS：PostgreSQL 17.11（Homebrew），同时进行了 MySQL 和 MariaDB 的安装与使用测试
数据库：tpch
数据库语言：SQL
实验工具：Homebrew、PostgreSQL psql、VS Code/Terminal
实验数据：TPC-H，Scale Factor = 0.01，采用实验提供的 CSV 数据文件</td></tr>
<tr><td colspan="4">实验结果及分析（给出 SQL 语句和相应的运行截图）
图片在下一页</td></tr>
</table>

```
(base) LSY@MacBook-Air-51 database % psql tpch
tpch=# SELECT p_partkey, p_name, p_brand, p_type, p_size
FROM part
WHERE p_brand LIKE 'Brand%'
  AND p_type LIKE '%POLISHED%'
ORDER BY p_size DESC;
 p_partkey |                    p_name                    |  p_brand   |          p_type
-----------+----------------------------------------------+------------+-------------------------
      1023 | grey blanched mint firebrick snow            | Brand#24   | PROMO POLISHED COPPER
      1583 | cream deep floral grey chiffon               | Brand#12   | ECONOMY POLISHED STEEL
       436 | turquoise yellow dim purple antique          | Brand#14   | LARGE POLISHED BRASS
      1979 | tomato forest pale moccasin magenta          | Brand#52   | SMALL POLISHED COPPER
       843 | frosted metallic mint lawn blanched          | Brand#15   | MEDIUM POLISHED STEEL
       767 | blush firebrick misty blanched purple        | Brand#24   | LARGE POLISHED TIN
      1333 | red sky dim light ivory                      | Brand#23   | MEDIUM POLISHED STEEL
      1572 | lemon midnight black papaya peru             | Brand#45   | ECONOMY POLISHED COPPER
      1753 | beige smoke slate steel lavender             | Brand#24   | ECONOMY POLISHED TIN
        90 | hot rosy violet plum pale                    | Brand#51   | ECONOMY POLISHED STEEL
        97 | coral dodger beige black chartreuse          | Brand#33   | MEDIUM POLISHED BRASS
      1863 | powder misty linen wheat coral               | Brand#54   | ECONOMY POLISHED BRASS
       635 | ghost aquamarine gainsboro lemon peach       | Brand#51   | PROMO POLISHED STEEL
      1670 | black tan metallic dark misty                | Brand#14   | MEDIUM POLISHED TIN
      1602 | mint red pale midnight orange                | Brand#51   | SMALL POLISHED STEEL
       940 | violet coral blanched orange light           | Brand#15   | PROMO POLISHED TIN
      1399 | brown lavender spring chocolate firebrick    | Brand#54   | ECONOMY POLISHED COPPER
       964 | sky orange indian pale burlywood             | Brand#54   | SMALL POLISHED BRASS
      1847 | yellow white puff orange rosy                | Brand#33   | SMALL POLISHED COPPER
        37 | royal coral orange burnished navajo          | Brand#45   | LARGE POLISHED TIN
        20 | ivory navy honeydew sandy midnight           | Brand#12   | LARGE POLISHED NICKEL
      1616 | moccasin tomato firebrick peru cyan          | Brand#33   | MEDIUM POLISHED TIN
       682 | lawn khaki green cornflower sienna           | Brand#53   | LARGE POLISHED NICKEL
       152 | white sky antique tomato chartreuse          | Brand#53   | MEDIUM POLISHED STEEL
       687 | frosted cornsilk tomato burnished smoke      | Brand#51   | LARGE POLISHED NICKEL
      1494 | blue violet white papaya royal               | Brand#15   | ECONOMY POLISHED TIN
       750 | puff slate magenta powder lawn               | Brand#43   | LARGE POLISHED NICKEL
       160 | frosted cornflower khaki salmon metallic     | Brand#55   | STANDARD POLISHED COPPEF
      1323 | burnished firebrick light steel lemon        | Brand#45   | SMALL POLISHED TIN
      1268 | tomato rosy beige yellow tan                 | Brand#31   | ECONOMY POLISHED STEEL
       274 | cyan lawn snow sky peru                      | Brand#23   | MEDIUM POLISHED STEEL
       224 | drab lavender moccasin almond purple         | Brand#13   | STANDARD POLISHED NICKEL
       795 | cream royal light yellow hot                 | Brand#54   | MEDIUM POLISHED TIN
      1539 | yellow orchid blue olive lavender            | Brand#54   | SMALL POLISHED TIN
      1178 | plum azure almond coral dim                  | Brand#24   | MEDIUM POLISHED TIN
       457 | navy linen forest moccasin slate             | Brand#52   | SMALL POLISHED TIN
      1060 | dark indian snow cornflower cream            | Brand#33   | STANDARD POLISHED BRASS
      1034 | lavender ghost sky beige navy                | Brand#52   | ECONOMY POLISHED BRASS
      1707 | misty plum sienna powder tan                 | Brand#51   | LARGE POLISHED NICKEL
       978 | cream tomato rose floral steel               | Brand#45   | MEDIUM POLISHED TIN
      1675 | antique orchid burlywood orange lime         | Brand#32   | SMALL POLISHED TIN
      1193 | olive purple turquoise cornflower honeydew   | Brand#11   | STANDARD POLISHED NICKEL
      1313 | floral turquoise navy cream lawn             | Brand#43   | ECONOMY POLISHED TIN
      1603 | saddle orange chocolate magenta turquoise    | Brand#43   | LARGE POLISHED TIN
        46 | honeydew turquoise aquamarine spring tan     | Brand#11   | STANDARD POLISHED TIN
      1617 | dodger grey drab moccasin lemon              | Brand#43   | LARGE POLISHED BRASS
      1098 | lace hot bisque firebrick lawn               | Brand#21   | ECONOMY POLISHED STEEL
      1487 | misty hot mint red bisque                    | Brand#45   | STANDARD POLISHED COPPEF
      1714 | brown lime sky purple salmon                 | Brand#31   | MEDIUM POLISHED TIN
```

p_partkey	p_name	p_brand	p_type
1023	grey blanched mint firebrick snow	Brand#24	PROMO POLISHED COPP
1583	cream deep floral grey chiffon	Brand#12	ECONOMY POLISHED ST
436	turquoise yellow dim purple antique	Brand#14	LARGE POLISHED BRAS
1979	tomato forest pale moccasin magenta	Brand#52	SMALL POLISHED COPP
843	frosted metallic mint lawn blanched	Brand#15	MEDIUM POLISHED STE
767	blush firebrick misty blanched purple	Brand#24	LARGE POLISHED TIN
1333	red sky dim light ivory	Brand#23	MEDIUM POLISHED STE
1572	lemon midnight black papaya peru	Brand#45	ECONOMY POLISHED CO
1753	beige smoke slate steel lavender	Brand#24	ECONOMY POLISHED TI
90	hot rosy violet plum pale	Brand#51	ECONOMY POLISHED ST
97	coral dodger beige black chartreuse	Brand#33	MEDIUM POLISHED BRA
1863	powder misty linen wheat coral	Brand#54	ECONOMY POLISHED BR
635	ghost aquamarine gainsboro lemon peach	Brand#51	PROMO POLISHED STEE
1670	black tan metallic dark misty	Brand#14	MEDIUM POLISHED TIN
1602	mint red pale midnight orange	Brand#51	SMALL POLISHED STEE
940	violet coral blanched orange light	Brand#15	PROMO POLISHED TIN
1399	brown lavender spring chocolate firebrick	Brand#54	ECONOMY POLISHED CO
964	sky orange indian pale burlywood	Brand#54	SMALL POLISHED BRAS
1847	yellow white puff orange rosy	Brand#33	SMALL POLISHED COPP
37	royal coral orange burnished navajo	Brand#45	LARGE POLISHED TIN
20	ivory navy honeydew sandy midnight	Brand#12	LARGE POLISHED NICK
1616	moccasin tomato firebrick peru cyan	Brand#33	MEDIUM POLISHED TIN
682	lawn khaki green cornflower sienna	Brand#53	LARGE POLISHED NICK
152	white sky antique tomato chartreuse	Brand#53	MEDIUM POLISHED STE
687	frosted cornsilk tomato burnished smoke	Brand#51	LARGE POLISHED NICK
1494	blue violet white papaya royal	Brand#15	ECONOMY POLISHED TI
750	puff slate magenta powder lawn	Brand#43	LARGE POLISHED NICK
160	frosted cornflower khaki salmon metallic	Brand#55	STANDARD POLISHED C
1323	burnished firebrick light steel lemon	Brand#45	SMALL POLISHED TIN
1268	tomato rosy beige yellow tan	Brand#31	ECONOMY POLISHED ST
274	cyan lawn snow sky peru	Brand#23	MEDIUM POLISHED STE
224	drab lavender moccasin almond purple	Brand#13	STANDARD POLISHED N
795	cream royal light yellow hot	Brand#54	MEDIUM POLISHED TIN
1539	yellow orchid blue olive lavender	Brand#54	SMALL POLISHED TIN
1178	plum azure almond coral dim	Brand#24	MEDIUM POLISHED TIN
457	navy linen forest moccasin slate	Brand#52	SMALL POLISHED TIN
1060	dark indian snow cornflower cream	Brand#33	STANDARD POLISHED B
1034	lavender ghost sky beige navy	Brand#52	ECONOMY POLISHED BR
1707	misty plum sienna powder tan	Brand#51	LARGE POLISHED NICK
978	cream tomato rose floral steel	Brand#45	MEDIUM POLISHED TIN
1675	antique orchid burlywood orange lime	Brand#32	SMALL POLISHED TIN
1193	olive purple turquoise cornflower honeydew	Brand#11	STANDARD POLISHED N
1313	floral turquoise navy cream lawn	Brand#43	ECONOMY POLISHED TI
1603	saddle orange chocolate magenta turquoise	Brand#43	LARGE POLISHED TIN
46	honeydew turquoise aquamarine spring tan	Brand#11	STANDARD POLISHED T
1617	dodger grey drab moccasin lemon	Brand#43	LARGE POLISHED BRAS
1098	lace hot bisque firebrick lawn	Brand#21	ECONOMY POLISHED ST
1487	misty hot mint red bisque	Brand#45	STANDARD POLISHED C
1714	brown lime sky purple salmon	Brand#31	MEDIUM POLISHED TIN
58	linen hot cornsilk drab bisque	Brand#53	STANDARD POLISHED T
942	coral peach bisque green seashell	Brand#35	PROMO POLISHED TIN
1483	navajo burlywood firebrick brown wheat	Brand#21	ECONOMY POLISHED ST
925	tomato antique plum snow burlywood	Brand#44	STANDARD POLISHED N
561	firebrick slate grey coral goldenrod	Brand#14	SMALL POLISHED STEE

:

```
(base) LSY@MacBook-Air-51 database % psql tpch
tpch=# SELECT c_mktsegment, COUNT(*) AS customer_count
FROM customer
GROUP BY c_mktsegment
ORDER BY c_mktsegment;
 c_mktsegment | customer_count
--------------+----------------
 AUTOMOBILE   |            302
 BUILDING     |            337
 FURNITURE    |            279
 HOUSEHOLD    |            294
 MACHINERY    |            288
(5 行记录)

tpch=# SELECT s_nationkey,
       ROUND(AVG(s_acctbal), 2) AS avg_acctbal
FROM supplier
GROUP BY s_nationkey
HAVING AVG(s_acctbal) > 9000
ORDER BY avg_acctbal DESC;
 s_nationkey | avg_acctbal
-------------+-------------
(0 行记录)

tpch=# SELECT *
FROM orders
ORDER BY o_totalprice DESC
LIMIT 10;
 o_orderkey | o_custkey | o_orderstatus | o_totalprice | o_orderdate | o_orderpriority |     o_clerk     | o_shippriority |                                o_com
------------+-----------+---------------+--------------+-------------+-----------------+-----------------+----------------+-----------------------------------------------------------------------------
      52965 |       676 | O             |    466001.28 | 1996-09-22  | 3-MEDIUM        | Clerk#000000103 |              0 | thless pinto beans use about the blithe
      29158 |       667 | O             |    439687.23 | 1995-10-21  | 2-HIGH          | Clerk#000000525 |              0 | lessly quiet deposits across the furiou
      44707 |      1013 | O             |    431771.98 | 1997-08-14  | 1-URGENT        | Clerk#000000313 |              0 | ly fluffily special ideas.
      59106 |       953 | O             |    430619.75 | 1996-10-24  | 4-NOT SPECIFIED | Clerk#000000344 |              0 | y above the slyly final deposits. furio
e s
       6882 |       178 | O             |    422359.65 | 1997-04-09  | 1-URGENT        | Clerk#000000974 |              0 | y dolphins. furiously ironic asymptotes
      57376 |       676 | O             |    411255.46 | 1995-06-20  | 1-URGENT        | Clerk#000000618 |              0 | ular Tiresias after the furiously regul
      39456 |      1016 | O             |    409770.83 | 1998-02-16  | 2-HIGH          | Clerk#000000972 |              0 | ts sleep final requests. carefully b
      17571 |       358 | F             |    408345.74 | 1992-03-16  | 4-NOT SPECIFIED | Clerk#000000503 |              0 | ckages. blithely silent requests use qu
 mainta
      39620 |      1279 | F             |    406938.36 | 1994-10-05  | 2-HIGH          | Clerk#000000457 |              0 | nic packages sleep quickly regular excu
lly! package
      35460 |       334 | F             |    405742.27 | 1993-11-28  | 5-LOW           | Clerk#000000112 |              0 |  deposits wake furiously into the s
(10 行记录)
```

```
(base) LSY@MacBook-Air-51 database % psql tpch
tpch=# SELECT l_orderkey,
       l_partkey,
       ROUND(l_extendedprice * (1 - l_discount), 2) AS actu
FROM lineitem
ORDER BY l_orderkey;
 l_orderkey | l_partkey | actual_amount
------------+-----------+---------------
          1 |      1552 |      23721.94
          1 |       674 |      51586.19
          1 |       637 |      11070.94
          1 |        22 |      23493.07
          1 |       241 |      24650.78
          1 |       157 |      31460.78
          2 |      1062 |      36596.28
          3 |        43 |      39890.59
          3 |       191 |      48121.48
          3 |      1285 |      30107.79
          3 |       294 |       2364.69
          3 |      1831 |      46578.47
          3 |       622 |      35629.31
          4 |       881 |      51852.71
          5 |      1086 |      14510.08
          5 |      1240 |      27595.18
          5 |       376 |      58713.02
          6 |      1397 |      44197.20
          7 |      1821 |      19226.67
          7 |      1453 |      11214.85
          7 |       948 |      76546.12
          7 |      1631 |      41626.23
          7 |      1519 |      49661.03
          7 |       793 |      55725.69
          7 |      1573 |       7077.94
         32 |       828 |      45986.61
         32 |      1980 |      59018.89
         32 |       442 |       2443.24
         32 |        28 |       3377.99
         32 |       859 |      73561.73
         32 |       117 |       5858.55
         33 |       614 |      42727.15
         33 |       606 |      47246.98
         33 |      1375 |       6062.76
         33 |       340 |      46277.09
         34 |       884 |      23203.44
         34 |       895 |      36348.81
         34 |      1696 |       9394.42
         35 |         5 |      21285.60
         35 |      1620 |      48630.98
         35 |      1209 |       7305.12
         35 |       852 |      41191.98
         35 |      1200 |      34445.54
         35 |       308 |      32817.43
         36 |      1198 |      42011.04
         37 |       227 |      41030.81
```

l_orderkey	l_partkey	actual_amount
1	1552	23721.94
1	674	51586.19
1	637	11070.94
1	22	23493.07
1	241	24650.78
1	157	31460.78
2	1062	36596.28
3	43	39890.59
3	191	48121.48
3	1285	30107.79
3	294	2364.69
3	1831	46578.47
3	622	35629.31
4	881	51852.71
5	1086	14510.08
5	1240	27595.18
5	376	58713.02
6	1397	44197.20
7	1821	19226.67
7	1453	11214.85
7	948	76546.12
7	1631	41626.23
7	1519	49661.03
7	793	55725.69
7	1573	7077.94
32	828	45986.61
32	1980	59018.89
32	442	2443.24
32	28	3377.99
32	859	73561.73
32	117	5858.55
33	614	42727.15
33	606	47246.98
33	1375	6062.76
33	340	46277.09
34	884	23203.44
34	895	36348.81
34	1696	9394.42
35	5	21285.60
35	1620	48630.98
35	1209	7305.12
35	852	41191.98
35	1200	34445.54
35	308	32817.43
36	1198	42011.04
37	227	41030.81
37	1268	43321.08
37	130	42080.81
38	1759	70150.08
39	24	36997.76
39	1866	42287.21

```
tpch=# SELECT p_container,
       COUNT(*) AS part_count
FROM part
WHERE p_container LIKE 'SM%'
GROUP BY p_container
ORDER BY p_container;
 p_container | part_count
-------------+------------
 SM BAG      |         58
 SM BOX      |         36
 SM CAN      |         50
 SM CASE     |         44
 SM DRUM     |         52
 SM JAR      |         54
 SM PACK     |         61
 SM PKG      |         59
(8 行记录)

tpch=# SELECT l_shipmode,
       ROUND(AVG(l_quantity), 2) AS avg_quantity
FROM lineitem
WHERE l_shipmode IN ('AIR', 'TRUCK')
GROUP BY l_shipmode
HAVING AVG(l_quantity) > 25;
 l_shipmode | avg_quantity
------------+--------------
 TRUCK      |        25.71
 AIR        |        25.48
(2 行记录)

tpch=# SELECT c_custkey,
       c_name,
       c_acctbal,
       ROUND(c_acctbal * 0.9, 2) AS discounted_balance
FROM customer
ORDER BY discounted_balance DESC
LIMIT 30;
 c_custkey |       c_name       | c_acctbal | discounted_bal
-----------+--------------------+-----------+---------------
       213 | Customer#000000213 |   9987.71 |            898
        45 | Customer#000000045 |   9983.38 |            898
      1106 | Customer#000001106 |   9977.62 |            897
       200 | Customer#000000200 |   9967.60 |            897
       140 | Customer#000000140 |   9963.15 |            896
       381 | Customer#000000381 |   9931.71 |            893
        43 | Customer#000000043 |   9904.28 |            891
       100 | Customer#000000100 |   9889.89 |            890
       780 | Customer#000000780 |   9874.12 |            888
       518 | Customer#000000518 |   9871.66 |            888
       197 | Customer#000000197 |   9860.22 |            887
       219 | Customer#000000219 |   9858.57 |            887
       987 | Customer#000000987 |   9850.64 |            886
```

```
tpch=# SELECT c_custkey,
       c_name,
       c_acctbal,
       ROUND(c_acctbal * 0.9, 2) AS discounted_balance
FROM customer
ORDER BY discounted_balance DESC;
 c_custkey |       c_name       | c_acctbal | discounted_
-----------+--------------------+-----------+------------
       213 | Customer#000000213 |   9987.71 |
        45 | Customer#000000045 |   9983.38 |
      1106 | Customer#000001106 |   9977.62 |
       200 | Customer#000000200 |   9967.60 |
       140 | Customer#000000140 |   9963.15 |
       381 | Customer#000000381 |   9931.71 |
        43 | Customer#000000043 |   9904.28 |
       100 | Customer#000000100 |   9889.89 |
       780 | Customer#000000780 |   9874.12 |
       518 | Customer#000000518 |   9871.66 |
       197 | Customer#000000197 |   9860.22 |
       219 | Customer#000000219 |   9858.57 |
       987 | Customer#000000987 |   9850.64 |
      1042 | Customer#000001042 |   9849.87 |
       468 | Customer#000000468 |   9834.19 |
      1370 | Customer#000001370 |   9802.04 |
      1479 | Customer#000001479 |   9793.29 |
       574 | Customer#000000574 |   9787.56 |
      1123 | Customer#000001123 |   9786.36 |
      1403 | Customer#000001403 |   9782.34 |
      1204 | Customer#000001204 |   9777.19 |
      1051 | Customer#000001051 |   9776.39 |
       919 | Customer#000000919 |   9774.97 |
       157 | Customer#000000157 |   9768.73 |
       601 | Customer#000000601 |   9768.21 |
       734 | Customer#000000734 |   9750.71 |
       487 | Customer#000000487 |   9749.37 |
       145 | Customer#000000145 |   9748.93 |
      1086 | Customer#000001086 |   9726.83 |
       592 | Customer#000000592 |   9712.75 |
      1478 | Customer#000001478 |   9701.54 |
      1026 | Customer#000001026 |   9699.28 |
       663 | Customer#000000663 |   9698.21 |
      1460 | Customer#000001460 |   9680.51 |
       606 | Customer#000000606 |   9676.98 |
       849 | Customer#000000849 |   9670.64 |
       279 | Customer#000000279 |   9663.23 |
       529 | Customer#000000529 |   9647.58 |
      1386 | Customer#000001386 |   9643.87 |
       810 | Customer#000000810 |   9632.77 |
       614 | Customer#000000614 |   9630.24 |
      1195 | Customer#000001195 |   9621.49 |
       768 | Customer#000000768 |   9618.84 |
       945 | Customer#000000945 |   9615.39 |
```

c_custkey	c_name	c_acctbal	discounted_
213	Customer#000000213	9987.71	
45	Customer#000000045	9983.38	
1106	Customer#000001106	9977.62	
200	Customer#000000200	9967.60	
140	Customer#000000140	9963.15	
381	Customer#000000381	9931.71	
43	Customer#000000043	9904.28	
100	Customer#000000100	9889.89	
780	Customer#000000780	9874.12	
518	Customer#000000518	9871.66	
197	Customer#000000197	9860.22	
219	Customer#000000219	9858.57	
987	Customer#000000987	9850.64	
1042	Customer#000001042	9849.87	
468	Customer#000000468	9834.19	
1370	Customer#000001370	9802.04	
1479	Customer#000001479	9793.29	
574	Customer#000000574	9787.56	
1123	Customer#000001123	9786.36	
1403	Customer#000001403	9782.34	
1204	Customer#000001204	9777.19	
1051	Customer#000001051	9776.39	
919	Customer#000000919	9774.97	
157	Customer#000000157	9768.73	
601	Customer#000000601	9768.21	
734	Customer#000000734	9750.71	
487	Customer#000000487	9749.37	
145	Customer#000000145	9748.93	
1086	Customer#000001086	9726.83	
592	Customer#000000592	9712.75	
1478	Customer#000001478	9701.54	
1026	Customer#000001026	9699.28	
663	Customer#000000663	9698.21	
1460	Customer#000001460	9680.51	
606	Customer#000000606	9676.98	
849	Customer#000000849	9670.64	
279	Customer#000000279	9663.23	
529	Customer#000000529	9647.58	
1386	Customer#000001386	9643.87	
810	Customer#000000810	9632.77	
614	Customer#000000614	9630.24	
1195	Customer#000001195	9621.49	
768	Customer#000000768	9618.84	
945	Customer#000000945	9615.39	
950	Customer#000000950	9609.77	
1022	Customer#000001022	9605.83	
1198	Customer#000001198	9593.35	
711	Customer#000000711	9591.51	
251	Customer#000000251	9585.32	
246	Customer#000000246	9584.96	
691	Customer#000000691	9566.15	

```
tpch=# SELECT s_nationkey,
       COUNT(*) AS supplier_count
FROM supplier
WHERE s_phone LIKE '15%'
GROUP BY s_nationkey
HAVING COUNT(*) > 5
ORDER BY supplier_count DESC;
 s_nationkey | supplier_count
-------------+----------------
(0 行记录)

tpch=# SELECT o_orderkey,
       o_custkey,
       o_orderdate,
       o_orderpriority,
       o_totalprice
FROM orders
WHERE o_orderpriority LIKE '%URGENT%'
ORDER BY o_orderdate ASC,
         o_totalprice DESC
LIMIT 15;
 o_orderkey | o_custkey | o_orderdate | o_orderpriority | o_totalpr
------------+-----------+-------------+-----------------+----------
       3271 |       331 | 1992-01-01  | 1-URGENT        |    138053
       1248 |       472 | 1992-01-02  | 1-URGENT        |    295111
      36582 |        34 | 1992-01-03  | 1-URGENT        |    199609
      47591 |       823 | 1992-01-03  | 1-URGENT        |    125040
      35811 |       586 | 1992-01-04  | 1-URGENT        |    185551
       9379 |       350 | 1992-01-04  | 1-URGENT        |    152004
      55011 |       833 | 1992-01-05  | 1-URGENT        |    214711
      31555 |       679 | 1992-01-05  | 1-URGENT        |    105491
      18625 |      1489 | 1992-01-06  | 1-URGENT        |    283425
      57698 |       953 | 1992-01-07  | 1-URGENT        |    189185
       2983 |       613 | 1992-01-07  | 1-URGENT        |     83588
      11296 |       832 | 1992-01-10  | 1-URGENT        |    378166
      34662 |       760 | 1992-01-11  | 1-URGENT        |    251035
      44033 |      1370 | 1992-01-11  | 1-URGENT        |     59313
      47879 |      1165 | 1992-01-12  | 1-URGENT        |     54960
(15 行记录)

tpch=# SELECT p_size,
       ROUND(AVG(p_retailprice), 2) AS avg_retailprice
FROM part
WHERE p_brand LIKE '_R%'
  AND p_mfgr LIKE 'Manufacturer%'
GROUP BY p_size
ORDER BY p_size;
 p_size | avg_retailprice
--------+-----------------
(0 行记录)
```

```
tpch=# SELECT EXTRACT(YEAR FROM l_shipdate) AS yea
       EXTRACT(QUARTER FROM l_shipdate) AS quarte
       COUNT(*) AS lineitem_count
FROM lineitem
WHERE l_shipdate >= DATE '1995-01-01'
  AND l_shipdate < DATE '1997-01-01'
GROUP BY EXTRACT(YEAR FROM l_shipdate),
         EXTRACT(QUARTER FROM l_shipdate)
HAVING COUNT(*) > 1000
ORDER BY year, quarter;
 year | quarter | lineitem_count
------+---------+----------------
 1995 |       1 |           2100
 1995 |       2 |           2135
 1995 |       3 |           2236
 1995 |       4 |           2302
 1996 |       1 |           2284
 1996 |       2 |           2284
 1996 |       3 |           2305
 1996 |       4 |           2327
(8 行记录)
```

2. 多表

```
tpch=# SELECT n.n_name,
       COUNT(c.c_custkey) AS customer_count
FROM nation n
LEFT JOIN customer c
  ON n.n_nationkey = c.c_nationkey
GROUP BY n.n_nationkey, n.n_name
ORDER BY n.n_name;
          n_name           | customer_count
---------------------------+----------------
 ALGERIA                   |             61
 ARGENTINA                 |             59
 BRAZIL                    |             68
 CANADA                    |             69
 CHINA                     |             58
 EGYPT                     |             66
 ETHIOPIA                  |             57
 FRANCE                    |             36
 GERMANY                   |             57
 INDIA                     |             60
 INDONESIA                 |             66
 IRAN                      |             72
 IRAQ                      |             58
 JAPAN                     |             67
 JORDAN                    |             54
 KENYA                     |             50
 MOROCCO                   |             72
 MOZAMBIQUE                |             62
 PERU                      |             56
 ROMANIA                   |             64
 RUSSIA                    |             59
 SAUDI ARABIA              |             67
 UNITED KINGDOM            |             56
 UNITED STATES             |             48
 VIETNAM                   |             58
(25 行记录)

tpch=# SELECT r.r_name,
       COUNT(s.s_suppkey) AS supplier_count
FROM region r
LEFT JOIN nation n
  ON r.r_regionkey = n.n_regionkey
LEFT JOIN supplier s
  ON n.n_nationkey = s.s_nationkey
GROUP BY r.r_regionkey, r.r_name
ORDER BY r.r_name;
          r_name           | supplier_count
---------------------------+----------------
 AFRICA                    |             21
 AMERICA                   |             20
 ASIA                      |             27
 EUROPE                    |             20
 MIDDLE EAST               |             12
(5 行记录)
```

```
tpch=# SELECT c.c_name,
       o.o_orderkey,
       o.o_totalprice
FROM customer c
JOIN orders o
  ON c.c_custkey = o.o_custkey
WHERE o.o_totalprice > 10000
ORDER BY o.o_totalprice DESC
LIMIT 30;
       c_name       | o_orderkey | o_totalprice
--------------------+------------+--------------
 Customer#000000676 |      52965 |    466001.28
 Customer#000000667 |      29158 |    439687.23
 Customer#000001013 |      44707 |    431771.98
 Customer#000000953 |      59106 |    430619.75
 Customer#000000178 |       6882 |    422359.65
 Customer#000000676 |      57376 |    411255.46
 Customer#000001016 |      39456 |    409770.83
 Customer#000000358 |      17571 |    408345.74
 Customer#000001279 |      39620 |    406938.36
 Customer#000000334 |      35460 |    405742.27
 Customer#000001003 |      15779 |    405401.76
 Customer#000000154 |      45382 |    405235.90
 Customer#000000583 |      52480 |    403464.01
 Customer#000000166 |      55937 |    402930.49
 Customer#000000085 |       4421 |    401055.62
 Customer#000001210 |      10209 |    400191.77
 Customer#000000772 |      35424 |    397797.80
 Customer#000001303 |      45670 |    397549.76
 Customer#000001096 |      45697 |    396261.24
 Customer#000000787 |      37216 |    395495.85
 Customer#000001375 |      11142 |    395039.05
 Customer#000001436 |      51975 |    393923.76
 Customer#000000424 |      48483 |    392614.07
 Customer#000000061 |      41445 |    392172.90
 Customer#000001147 |      54819 |    386714.93
 Customer#000000379 |      24832 |    386334.55
 Customer#000000458 |      14179 |    384265.43
 Customer#000001318 |      43142 |    382233.07
 Customer#000000665 |      40132 |    381793.12
 Customer#000000805 |      27746 |    381170.13
(30 行记录)
```

```
tpch=# SELECT *
FROM customer c
WHERE EXISTS (
    SELECT 1
    FROM orders o
    WHERE o.o_custkey = c.c_custkey
)
ORDER BY c.c_custkey
LIMIT 30;
 c_custkey |       c_name       |                c_address                | c_nationkey |     c_phone     | c_acctbal | c_mktsegment |
mment
-----------+--------------------+-----------------------------------------+-------------+-----------------+-----------+--------------+-----------------------------------------------------------------------------
-----------------------------------------
         1 | Customer#000000001 | IVhzIApeRb ot,c,E                       |          15 | 25-989-741-2988 |    711.56 | BUILDING     | to the even, regular platelets
 nag e
         2 | Customer#000000002 | XSTf4,NCwDVaWNe6tEgvwfmRchLXak          |          13 | 23-768-687-3665 |    121.65 | AUTOMOBILE   | l accounts. blithely ironic th
: caref
         4 | Customer#000000004 | XxVSJsLAGtn                             |           4 | 14-128-190-5944 |   2866.83 | MACHINERY    |  requests. final, regular idea
         5 | Customer#000000005 | KvpyuHCplrB84WgAiGV6sYpZq7Tj            |           3 | 13-750-942-6364 |    794.47 | HOUSEHOLD    | n accounts will have to unwind
         7 | Customer#000000007 | TcGe5gaZNgVePxU5kRrvXBfkasDTea          |          18 | 28-190-982-9759 |   9561.95 | AUTOMOBILE   | ainst the ironic, express theo
to beans among the exp
         8 | Customer#000000008 | I0B10bB0AymmC, 0PrRYBCP1yGJ8xcBPmWhl5   |          17 | 27-147-574-9335 |   6819.74 | BUILDING     | among the slyly regular theodo
ts. carefully even theodolites haggle slyly along the ide
        10 | Customer#000000010 | 6LrEaV6KR6PLVcgl2ArL Q3rqzLzcT1 v2      |           5 | 15-741-346-9870 |   2753.54 | HOUSEHOLD    | es regular deposits haggle. fu
        11 | Customer#000000011 | PkWS 3HlXqwTuzrKg633BEi                 |          23 | 33-464-151-3439 |   -272.60 | BUILDING     | ckages. requests sleep slyly.
romise above the slyly regular pinto beans.
        13 | Customer#000000013 | nsXQu0oVjD7PM659uC3SRSp                 |           3 | 13-761-547-5974 |   3857.34 | BUILDING     | ounts sleep carefully after th
old notornis use ironic requests. blithely
        14 | Customer#000000014 | KXkletMlL2JQEA                          |           1 | 11-845-129-3851 |   5266.30 | FURNITURE    | , ironic packages across the u
        16 | Customer#000000016 | cYiaeMLZSMAOQ2 d0W,                     |          10 | 20-781-609-3107 |   4681.03 | FURNITURE    | kly silent courts. thinly regu
fily after
        17 | Customer#000000017 | izrh 6jdqtp2eqdtbkswDD8SG4SzXruMfIXyR7 |           2 | 12-970-682-3487 |      6.34 | AUTOMOBILE   | packages wake! blithely even p
        19 | Customer#000000019 | uc,3bHIx84H,wdrmLOjVsiqXCq2tr           |          18 | 28-396-526-5053 |   8914.71 | HOUSEHOLD    |  nag. furiously careful packag
ts. furiously regular in
        20 | Customer#000000020 | JrPk8Pqplj4Ne                           |          22 | 32-957-234-8742 |   7603.40 | FURNITURE    | g alongside of the special exc
ackages wake
        22 | Customer#000000022 | QI6p41,FNs5k7RZoCCVPUTkUdYpB            |           3 | 13-806-545-9701 |    591.98 | MACHINERY    | s nod furiously above the furi
        23 | Customer#000000023 | OdY W13N7Be3OC5MpgfmcYss0Wn6TKT         |           3 | 13-312-472-8245 |   3332.02 | HOUSEHOLD    | deposits. special deposits caj
l deposits about the furiously
        25 | Customer#000000025 | Hp8GyFQgGHFYSilH5tBfe                   |          12 | 22-603-468-3533 |   7133.70 | FURNITURE    | y. accounts sleep ruthlessly a
eodolites. unusual instructions sleep. ironic, final
        26 | Customer#000000026 | 8ljrc5ZeMl7UciP                         |          22 | 32-363-455-4837 |   5182.05 | AUTOMOBILE   | c requests use furiously ironi
ependencies us
        28 | Customer#000000028 | iVyg0daQ,Tha8x2WPWA9m2529m              |           8 | 18-774-241-1462 |   1007.18 | FURNITURE    |  along the regular deposits. f
        29 | Customer#000000029 | sJ5adtfyAkCK63df2,vF25zyQMVYE34uh       |           0 | 10-773-203-7342 |   7618.27 | FURNITURE    | its after the carefully final
        31 | Customer#000000031 | LUACbO0viaAv6eXOAebryDB xjVst           |          23 | 33-197-837-7094 |   5236.89 | HOUSEHOLD    | s use among the blithely pendi
        32 | Customer#000000032 | jD2xZzi UmId,DCtNBLXKj9q0Tlp2iQ6ZcO3J   |          15 | 25-430-914-2194 |   3471.53 | BUILDING     | cial ideas. final, furious req
        34 | Customer#000000034 | Q6G9wZ6dnczmtOx509xgE,M2KV              |          15 | 25-344-968-5422 |   8589.70 | HOUSEHOLD    | nder against the even, pending
        35 | Customer#000000035 | TEjWGE4nBzJL2                           |          17 | 27-566-888-7431 |   1228.24 | HOUSEHOLD    | requests. special, express req
        37 | Customer#000000037 | 7EV4Pwh,3SboctTWt                       |           8 | 18-385-235-7162 |   -917.75 | FURNITURE    | ilent packages are carefully a
l
        38 | Customer#000000038 | a5Ee5e9568R8RLP 2ap7                    |          12 | 22-306-880-7212 |   6345.11 | HOUSEHOLD    | lar excuses. closely even asym
uses. carefully silent pinto beans sleep carefully fin
        40 | Customer#000000040 | gOnGWAyhSV1ofv                          |           3 | 13-652-915-8939 |   1335.30 | BUILDING     | rges impress after the slyly i
lithely
        41 | Customer#000000041 | IM9mzmyoxeBmvNw8lA7G3Ydska2nkZF         |          10 | 20-917-711-4011 |    270.95 | HOUSEHOLD    | ly regular accounts hang bold,
```

问题　输出　调试控制台　终端　端口

```
        40 | Customer#000000040 | gOnGWAyhSV1ofv                          |           3 | 13-652-915-8939 |   1335.30 | BUILDING     | rges impress after the slyly ir
lithely
        41 | Customer#000000041 | IM9mzmyoxeBmvNw8lA7G3Ydska2nkZF         |          10 | 20-917-711-4011 |    270.95 | HOUSEHOLD    | ly regular accounts hang bold,
foxes haggle slyly above the special, final depo
        43 | Customer#000000043 | ouSbjHk8lh5fKX3zGso3ZSIj9Aa3PoaFd       |          19 | 29-316-665-2897 |   9804.28 | MACHINERY    | ial requests: carefully pending
refully final courts cajole quickly. carefully
        44 | Customer#000000044 | Oi,dOSPwDu4jo4x,,P85E0dmhZGvNtBwi       |          16 | 26-190-260-5375 |   7315.94 | AUTOMOBILE   | r requests around the unusual,
(30 行记录)

tpch=# SELECT n.n_name,
       ROUND(AVG(c.c_acctbal), 2) AS avg_acctbal
FROM nation n
JOIN customer c
  ON n.n_nationkey = c.c_nationkey
GROUP BY n.n_nationkey, n.n_name
ORDER BY n.n_name;
          n_name          | avg_acctbal
--------------------------+-------------
 ALGERIA                  |     4068.53
 ARGENTINA                |     4850.90
 BRAZIL                   |     3635.30
 CANADA                   |     4116.12
 CHINA                    |     5032.12
 EGYPT                    |     4128.49
 ETHIOPIA                 |     3539.67
 FRANCE                   |     3907.31
 GERMANY                  |     4280.10
 INDIA                    |     4566.70
 INDONESIA                |     4971.41
 IRAN                     |     4206.76
 IRAQ                     |     4613.16
 JAPAN                    |     4962.46
 JORDAN                   |     4186.00
 KENYA                    |     4901.11
 MOROCCO                  |     5484.47
 MOZAMBIQUE               |     4584.81
 PERU                     |     4301.28
 ROMANIA                  |     3941.04
 RUSSIA                   |     4321.54
 SAUDI ARABIA             |     5495.69
 UNITED KINGDOM           |     3028.29
 UNITED STATES            |     4297.54
 VIETNAM                  |     4712.10
(25 行记录)
```

```
tpch=# SELECT *
FROM customer c
WHERE NOT EXISTS (
    SELECT 1
    FROM orders o
    WHERE o.o_custkey = c.c_custkey
)
ORDER BY c.c_custkey;
 c_custkey |       c_name       |                c_address                | c_nationkey |     c_phone     | c_acctbal | c_mktsegment |
_comment
-----------+--------------------+-----------------------------------------+-------------+-----------------+-----------+--------------+-----------
-----------------------------------------
         3 | Customer#000000003 | MG9kdTD2WBHm                            |           1 | 11-719-748-3364 |   7498.12 | AUTOMOBILE   |  deposits eat slyly ironic,
 foxes detect slyly. blithely even accounts abov
         6 | Customer#000000006 | sKZz0CsnMD7mp4Xd0YrBvx,LREYKUWAh yVn    |          20 | 30-114-968-4951 |   7638.57 | AUTOMOBILE   | tions. even deposits boost
 packages. final accounts cajole requests. furious
         9 | Customer#000000009 | xKiAFTjUsCuxfeleNqefumTrjS              |           8 | 18-338-906-3675 |   8324.07 | FURNITURE    | r theodolites according to
cuses: pending requests haggle furiousl
        12 | Customer#000000012 | 9PWKuhzT4Zr1Q                           |          13 | 23-791-276-1263 |   3396.49 | HOUSEHOLD    |  to the carefully final bra
sts nag. ironic theodolites boost quickly along
        15 | Customer#000000015 | YtWggXoOLdwdo7b0y,BZaGUQMLJMX1Y,EC,6Dn  |          23 | 33-687-542-7601 |   2788.52 | HOUSEHOLD    |  platelets. regular deposit
ly unusual packages nag slyly at the fluf
        18 | Customer#000000018 | 3txGO AiuFux3zT0Z9NYaFRnZt              |           6 | 16-155-215-1315 |   5494.43 | BUILDING     | s sleep. carefully even ins
ngside of t
        21 | Customer#000000021 | XYmVpr9yAHDEn                           |           8 | 18-902-614-8344 |   1428.25 | MACHINERY    |  quickly final accounts int
        24 | Customer#000000024 | HXAFgIAyjxtdqwimt13Y3OZO 4xeLe7U8PqG    |          13 | 23-127-851-8031 |   9255.67 | MACHINERY    | into beans. fluffily final
        27 | Customer#000000027 | IS8GIyxpBrLpMT0u7                       |           3 | 13-137-193-2709 |   5679.84 | BUILDING     |  about the carefully ironic
        30 | Customer#000000030 | nJDsELGAavU63Jl0c5NKsKfL8rIJQQkQnYL2QJY |           1 | 11-764-165-5076 |   9321.01 | BUILDING     | lithely final requests. fur
        33 | Customer#000000033 | qFSlMuLucBmx9xnn5ib2csWUweg D           |          17 | 27-375-391-1280 |    -78.56 | AUTOMOBILE   | s. slyly regular accounts a
ding requests
        36 | Customer#000000036 | 3TvCzjuPzpJ0,DdJ8kW5U                   |          21 | 31-704-669-5769 |   4987.27 | BUILDING     | haggle. enticing, quiet pla
eaves. carefully regular acc
        39 | Customer#000000039 | nnbRg,Pvy33dfkorYE FdeZ60               |           2 | 12-387-467-6509 |   6264.31 | AUTOMOBILE   | tions. slyly silent excuses
        42 | Customer#000000042 | ziSrvyyBke                              |           5 | 15-416-330-4175 |   8727.01 | BUILDING     | ssly according to the pinto
equests across the even, pending accounts wake special
        45 | Customer#000000045 | 4v3OcpFgoOmMG,CbnF,4mdC                 |           9 | 19-715-298-9917 |   9983.38 | AUTOMOBILE   | nto beans haggle slyly alon
        48 | Customer#000000048 | 0UU iPhBupFvemNB                        |           0 | 10-508-348-5882 |   3792.50 | BUILDING     | re fluffily pending foxes.
ep slyly. even platelets cajo
        51 | Customer#000000051 | uR,wEaiTvo4                             |          12 | 22-344-885-4251 |    855.87 | FURNITURE    | eposits. furiously regular
y packages. furious
        54 | Customer#000000054 | ,k4vf 5vECGWFy,hosTE,                   |           4 | 14-776-370-4745 |    868.90 | AUTOMOBILE   | sual, silent accounts. furi
le special deposits. final, final accounts use furi
        57 | Customer#000000057 | 97XYbsuOPRXPWU                          |          21 | 31-835-306-1
650 |   4151.93 | AUTOMOBILE   | ove the carefully special packages. even, unusual deposits sleep slyly
 pend
        60 | Customer#000000060 | FyodhjwMChsZmUz7Jz0H                    |          12 | 22-480-575-5
866 |   2741.87 | MACHINERY    | latelets. blithely unusual courts boost furiously about the packages.
blithely final instruct
        63 | Customer#000000063 | IXRSpVWWZraKII                          |          21 | 31-952-552-9
```

```
tpch=# SELECT n.n_name,
       c1.c_custkey AS customer1_id,
       c1.c_name AS customer1_name,
       c2.c_custkey AS customer2_id,
       c2.c_name AS customer2_name
FROM customer c1
JOIN customer c2
  ON c1.c_nationkey = c2.c_nationkey
 AND c1.c_custkey < c2.c_custkey
JOIN nation n
  ON c1.c_nationkey = n.n_nationkey
ORDER BY n.n_name, customer1_id, customer2_id
LIMIT 30;
          n_name           | customer1_id |   customer1_name   | customer2_id |   custome
---------------------------+--------------+--------------------+--------------+----------
 ALGERIA                   |           29 | Customer#000000029 |           48 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |           73 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |           76 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |           80 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |           86 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          155 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          165 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          266 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          295 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          304 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          313 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          318 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          337 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          456 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          491 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          541 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          579 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          618 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          643 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          659 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          660 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          662 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          700 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          751 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          754 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          790 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          793 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          797 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          802 | Customer#
 ALGERIA                   |           29 | Customer#000000029 |          817 | Customer#
(30 行记录)
```

```
tpch=# SELECT o.o_orderkey,
       o.o_custkey,
       o.o_totalprice
FROM orders o
WHERE o.o_totalprice > (
    SELECT AVG(o2.o_totalprice)
    FROM orders o2
    WHERE o2.o_custkey = o.o_custkey
)
ORDER BY o.o_custkey, o.o_totalprice DESC
LIMIT 30;
 o_orderkey | o_custkey | o_totalprice
------------+-----------+--------------
       9154 |         1 |    357345.46
      36422 |         1 |    270087.44
      24322 |         1 |    231040.44
      53283 |         1 |    165928.33
      38276 |         2 |    201568.55
      20257 |         2 |    181875.60
       6980 |         2 |    170842.93
      29408 |         2 |    154867.09
      10563 |         2 |    143707.70
      29956 |         2 |    116247.57
      26407 |         4 |    275581.07
      58146 |         4 |    241263.01
        739 |         4 |    226008.80
      59140 |         4 |    215749.35
      28417 |         4 |    209874.16
      54722 |         4 |    208672.53
      56196 |         4 |    199826.92
      41600 |         4 |    192516.42
      24416 |         4 |    184512.04
      54404 |         4 |    182306.44
      26722 |         4 |    173562.06
      14947 |         4 |    165118.45
      34755 |         4 |    156681.56
      25893 |         4 |    150695.21
      38117 |         4 |    148782.33
      10788 |         4 |    147767.08
       7141 |         5 |    271505.25
      11746 |         5 |    177360.54
      11682 |         5 |    153588.74
      14308 |         5 |    132817.36
(30 行记录)
```

```
tpch=# SELECT s.s_suppkey,
       s.s_name
FROM supplier s
WHERE NOT EXISTS (
    SELECT DISTINCT p.p_type
    FROM part p
    WHERE NOT EXISTS (
        SELECT 1
        FROM partsupp ps
        JOIN part p2
          ON ps.ps_partkey = p2.p_partkey
        WHERE ps.ps_suppkey = s.s_suppkey
          AND p2.p_type = p.p_type
    )
);
 s_suppkey | s_name
-----------+--------
(0 行记录)

tpch=# SELECT o.o_orderkey,
       o.o_orderdate,
       o.o_totalprice
FROM orders o
WHERE o.o_totalprice > (
    SELECT AVG(o2.o_totalprice)
    FROM orders o2
    WHERE EXTRACT(YEAR FROM o2.o_orderdate)
          = EXTRACT(YEAR FROM o.o_orderdate)
      AND EXTRACT(MONTH FROM o2.o_orderdate)
          = EXTRACT(MONTH FROM o.o_orderdate)
)
ORDER BY o.o_orderdate, o.o_totalprice DESC
LIMIT 30;

 o_orderkey | o_orderdate | o_totalprice
------------+-------------+--------------
      45697 | 1992-01-01  |    396261.24
      27137 | 1992-01-01  |    258661.28
      37543 | 1992-01-01  |    168013.86
      51648 | 1992-01-02  |    306254.24
       1248 | 1992-01-02  |    295111.15
       3712 | 1992-01-02  |    181818.96
      22693 | 1992-01-03  |    246087.75
      36582 | 1992-01-03  |    199609.11
       7618 | 1992-01-04  |    297074.70
      35811 | 1992-01-04  |    185551.43
      10945 | 1992-01-04  |    168422.71
       9379 | 1992-01-04  |    152004.78
       7107 | 1992-01-05  |    286134.31
      55011 | 1992-01-05  |    214711.38
       6278 | 1992-01-05  |    160399.31
      13444 | 1992-01-06  |    304179.27
      18625 | 1992-01-06  |    283425.83
      16036 | 1992-01-06  |    256206.81
       5601 | 1992-01-06  |    175212.24
      10432 | 1992-01-06  |    167753.06
```

```
tpch=# SELECT o.o_orderkey,
       o.o_orderdate,
       o.o_totalprice
FROM orders o
WHERE o.o_totalprice > (
    SELECT AVG(o2.o_totalprice)
    FROM orders o2
    WHERE EXTRACT(YEAR FROM o2.o_orderdate)
          = EXTRACT(YEAR FROM o.o_orderdate)
      AND EXTRACT(MONTH FROM o2.o_orderdate)
          = EXTRACT(MONTH FROM o.o_orderdate)
)
ORDER BY o.o_orderdate, o.o_totalprice DESC
LIMIT 30;

 o_orderkey | o_orderdate | o_totalprice
------------+-------------+--------------
      45697 | 1992-01-01  |    396261.24
      27137 | 1992-01-01  |    258661.28
      37543 | 1992-01-01  |    168013.86
      51648 | 1992-01-02  |    306254.24
       1248 | 1992-01-02  |    295111.15
       3712 | 1992-01-02  |    181818.96
      22693 | 1992-01-03  |    246087.75
      36582 | 1992-01-03  |    199609.11
       7618 | 1992-01-04  |    297074.70
      35811 | 1992-01-04  |    185551.43
      10945 | 1992-01-04  |    168422.71
       9379 | 1992-01-04  |    152004.78
       7107 | 1992-01-05  |    286134.31
      55011 | 1992-01-05  |    214711.38
       6278 | 1992-01-05  |    160399.31
      13444 | 1992-01-06  |    304179.27
      18625 | 1992-01-06  |    283425.83
      16036 | 1992-01-06  |    256206.81
       5601 | 1992-01-06  |    175212.24
      10432 | 1992-01-06  |    167753.06
      52516 | 1992-01-06  |    161783.91
      56645 | 1992-01-07  |    277027.85
      57698 | 1992-01-07  |    189185.83
      54115 | 1992-01-08  |    190349.91
      26818 | 1992-01-09  |    310219.64
      54307 | 1992-01-09  |    209670.97
       7042 | 1992-01-09  |    208635.44
       5409 | 1992-01-09  |    190710.92
       4292 | 1992-01-09  |    177630.99
      59456 | 1992-01-09  |    162010.30
(30 行记录)
```

实验总结

本实验基于 TPC-H 数据库完成了单表查询和多表查询，掌握了条件筛选、分组统计、排序、聚合函数、连接查询及子查询等 SQL 操作，加深了对 SQL 查询语句及数据库表间关系的理解。